DE–7853 **11**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, MAY 2014.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Convert decimal 41 to binary. (4)

   (b) Using 10's complement, subtract 72532–3250. (4)

2. Discuss on number and character codes with examples.

3. Simplify the Boolean function:

   $F = A'C + A'B + AB'C + BC$.

4. Design a BCD–to–excess–3 code converter.

5. Draw the logic diagram and function table for arithmetic circuit.

6. Summarize the procedure of master slave flip–flop.

7. Give a brief account on flip–flop excitation tables.

8. Elucidate the role of control unit with respect to instruction and execution cycle.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Convert $(0.513)_{10}$ to octal (5)

(b) Convert $(306.D)_{16}$ to binary (5)

(c) Convert $(10.375)_{10}$ to octal (5)

10. Illustrate and explain the concepts of multiplexers and demultiplexers.

11. Simplify the following Boolean functions in

(a) Sum of products

(b) Product of sums

$$F(A,B,C,D) = \sum(0,1,2,5,8,9,10).$$

12. Illustrate and explain the half adder and full adder circuits.

13. Give a neat description on memory unit with neat diagram.

14. Elaborate the ripple counters with neat diagrams.

15. Describe the concept direct memory access.

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DE–7854 **12**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, MAY 2014.

First Semester

C AND DATA STRUCTURES

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. List down the any four types of operators and expressions with examples.

2. Illustrate and explain the branching statement, "If–Else".

3. Write a C program to find whether the given string is palindrome or not.

4. Discuss the concept of recursion with a simple program.

5. Narrate the elements of defining a function. Also discuss with examples.

6. Develop a C program that uses the functions ftell and fseek.

7. List and explain the various types of queue operations.

8. Give a note on binary tree and its applications with neat diagrams wherever necessary.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. With syntax and examples, explain the input and output statements, both formatted and unformatted categories.

10. Develop a C program to find the sum of above diagonal, below diagonal and trace of a square matrix, using two–dimensional arrays concept.

11. Write a program to find the sum of first n odd numbers using any one of the control statement.

12. Develop a C program using pointers to compute the sum of all elements stored in an array.

13. Write a C program to read data from the keyboard, write it to a file called INPUT, again read the same data from the INPUT file and display it on the screen.

14. Give a neat description on stack structure. Illustrate and explain the types of operations applied to a stack.

15. Compare and contrast singly linked list with doubly linked list with an example.

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DE–7855 **13**

DISTANCE EDUCATION

P.G.D.C.A (S) EXAMINATION, MAY 2014.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. How data base differs from a data file?

2. Explain the concept of schema.

3. Discuss on the semantic issues.

4. Mention the data manipulation facilities.

5. List the benefits of Oracle.

6. What is meant by physical file?

7. How to join multiple tables using Query?

8. Discuss on embedded SQL statements.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the advantages of database management system.

10. Explain the concept of user interfaces in DBMS.

11. List the concepts of relational model.

12. Explain the following :

    (a) Data definition facilities

    (b) Data control facilities

13. Describe the architecture of Oracle database.

14. Discuss on any five DML commands with examples.

15. How to create, delete, truncate and drop a table? Explain.

DE–7856 **14**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

First Semester

OPERATING SYSTEMS

(2011 Calendar Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions in about 200 words each.

1. What is an Operating System? Explain the system calls.

2. What do you mean by Semaphores? Explain in detail.

3. Explain the concept of multiple queues.

4. What are Device Controllers? Explain.

5. What is deadlock? How to avoid it?

6. Discuss the concept of memory management with Bit-maps.

7. What is meant by segmentation? Explain how sharing is accomplished in a pure segmentation system.

8. Discuss about Disk space management.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions in about 400 words each.

9. Discuss about the history of operating systems.

10. Explain the following :

    (a) Race conditions

    (b) Mutual exclusion.

11. Describe neatly about the round robin scheduling.

12. What are device drivers? Explain about the Device-Independent I/O software.

13. Explain the concept of variable partition multiprogramming with diagrams.

14. What do you mean by paging? Describe the page replacement algorithm.

15. Discuss about file servers and security.

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DE–7857 **21**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, MAY 2014.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Discuss the paradigm for object oriented programming.

2. How variables are defined and classified in C++?

3. Explain the execution of IF statement.

4. Define structures. Discuss its concept with examples.

5. Explain the concept of friend function.

6. Discuss the usage of macros in C++.

7. Define constructors. Discuss its types.

8. How to manage the errors during file operations?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Mention the basic concepts of object oriented programming.

10. Write a program in C++ to find the factorial of an integer using recursion function.

11. How WHILE statement performs its execution?

12. Write a program in C++ to arrange a set of numbers in ascending order.

13. Explain the concept of argument passing through value and reference.

14. Write a program in C++ to implement the concept of operator overloading.

15. Discuss on command line arguments.

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DE–7858 | **22**

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, MAY 2014.

Second Semester

UNIX AND SHELL PROGRAMMING

(2011 Calendar year onwards)

Time : 3 Hours Maximum : 100 Marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss about the directories and filenames in Unix.

2. Brief about command arguments and parameters.

3. Explain the while and until loops in detail.

4. Describe the standard input and output in Unix.

5. Write a note on Built in Functions.

6. Explain about the Unix system calls.

7. Brief in detail about functions and procedures.

8. Discuss about the filters in Unix.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe in detail about awk pattern scanning.

10. Write a shell program to display prime numbers between 1 to 100.

11. Write a shell script to sorting given numbers in descending order.

12. Brief all the IS commands in detail.

13. Discuss in detail about signal and interrupts in Unix.

14. Write a shell script program to convert a binary number into its decimal equivalent.

15. (a) Discuss in detail about the directories and inodes in Unix.

    (b) Write a note on macro in Unix.

DE–7859 **23**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

Second Semester

COMMUNICATION SKILLS

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss about :
   (a) Communication concepts and Definition
   (b) Types of communication.

2. Short notes on :
   (a) Reading styles
   (b) Report writing and minutes.

3. Discuss about :
   (a) Methods of asking questions
   (b) Improving speech delivery.

4. Write short notes on :
   (a) Personality test through Group discussion
   (b) Body language as a powerful communication.

5. Write about Negotiation techniques in detail.

6. Explain about :
   (a) Semantics
   (b) Downward and upward communication.

7. Discuss about :

   (a) The Union role in Communication

   (b) Employee Communication.

8. Explain about :

   (a) Techniques for delivering presentation

   (b) A post script.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about effective writing skills in detail.

10. Explain in detail about speaking techniques.

11. Write an essay about Group discussion and various techniques.

12. Explain about Negotiation processing and mock negotiation.

13. Explain in detail about meaning and participating in meetings and seminars.

14. Write an essay about communication management and its relationship in detail.

15. Discuss briefly the presentation details.

DE–7860 **24**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

Second Semester

COMPUTER GRAPHICS

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various input/output devices.

2. Describe the simple DDA algorithm for drawing a line.

3. Mention the principles involved in two dimensional transformations.

4. How windows and view ports are defined?

5. Explain the concept of scaling in three dimensional orientation.

6. What is meant by Projection?

7. Discuss on Z–Buffer algorithm.

8. Give the importance of Feedback.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on any two applications of computer graphics.

10. How to draw a ellipse through graphics?

11. Describe the algorithm for clipping the convex polygon.

12. Give the matrix representation for rotating a two dimensional object.

    (a) through the origin

    (b) through any arbitrary point.

13. How to construct a three dimensional view?

14. Describe the algorithm for hidden surface problem.

15. Give the importance of the command language.

DE–8785 **15**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

First Semester

DATA STRUCTURES USING C – LAB – I

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C program to the factorial of a given number using recursion.

   (b) Write a C program to implement insert and delete operations on Queue using array concept.

---------- Cut here ----------

2. (a) Write C program to copy contents of one file to another file

   (b) Write a C program for linked list implementation of Queue operations.

---------- Cut here ----------

3. (a) Write a C program to read 10 values to an array variable. Use pointer to locate and display each value.

   (b) Write a C program to evaluate the mathematical expression using stack.

4. (a) Write a C program to find the all possible roots of a quadratic equation using function $ax^2 + bx + c = 0$.

   (b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

5. (a) Write a C program to multiply any two order of matrix using function.

   (b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

6. (a) A man is paid at the hourly rate of Rs. 15/- per hour for the first 45 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for further hours worked. Write a C program to input the no. of hours worked per week, calculation and print his gross weekly wage.

   (b) Write a C program for linked list implementation of Queue operations.

---------- Cut here ----------

7. (a) Write a C program to read 10 values to an array variable. Use pointer to locate and display each value.

   (b) Write a C program to evaluate the mathematical expression using stack.

---------- Cut here ----------

8. (a) Write a C program to the factorial of a given number using recursion

   (b) Write a C program to implement insert and delete operations on Queue using array concept.

DE-8786 **16**

DISTANCE EDUCATION

P.G.D.C.A. (S) DEGREE EXAMINATION, MAY 2014.

First Semester

Lab II – RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. Create a table product with the following fields :
Product id, product name, rate, qty. Also write down the statements for the following.

   (a) Insert at least 7 records on it.

   (b) Count the number of product whose rate is greater than Rs.100.

   (c) Count the number of product whose rate between Rs.50 to Rs.150.

   (d) List down the product details whose product name starts with P.

2. Write a program in SQL for hospital billing system with the following fields.

   Patient no, Patient name, Age, Attending Doctor, Patient-type (IN/OUT), consulting charge, blood test charge,
   X-Ray charges, other test charges and total fee.

   Prepare reports with the following conditions :

   (a) The patients, undergone blood test.

   (b) The patients, who belong to out-patient category.

   (c) The patients, who taken X-rays.

   The report should included the details such as patient no, patient name, age, charges and total charges.

---------- Cut here ----------

3. Write a PL/SQL program to prepare a mark statement of a student using student table and mark table. Assume your own data.

---------- Cut here ----------

4. Create a Airline reservation table with the following fields, flight no, class, passenger name, seat number and

   Personal table with following fields passenger name, sex, age, marital status, nationality. Write a program in ORACLE to prepare the following reports :

   (a) List the passenger names with seat number according to class and flight numbers.

   (b) Total number of married female passengers in a particular flight.

   (c) List out the male passengers in the age between 18 to 25 for all flights.

5. A company ABC Pvt Ltd., states monthly salary to its employee. It consists of basic salary, allowance, deduction.

D.A. – 43% of Basic pay

H.R.A. — 7% of (Basic pay +D.A)

Deductions :

P.F. – subscribed by an capital.

LIC premium – payable by an employee, the salary saving scheme.

Loan Recovery – if any payable by an employee.

Create a master table of 10 records with following fields.

Employee No, name, designation, basic pay, HRA, DA, Bank A/c No., LIC premium no.,

A transaction file contains :

Employee no, P.F. subscription, LIC premium amount, loan recovery.

Write a program in SQL to prepare a report with following information.

Serial no., Employee no., Bank A/c no., Name, Basic Pay, Total Allowances, Gross Pay, Total deductions, Netpay.

DE–8787 **25**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

Second Semester

LAB III — GRAPHICS LAB USING C++

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Break-Up of Marks :

Record	:	10
Algorithm	:	10
Program	:	50
Debugging	:	10
Execution	:	10
Result	:	10
Total	:	100

1. Create a base class shape, which contains the common elements of circle, triangle and rectangle, from which you derive three classes called ball, rect and tria. Include two member functions set ( ) to set coordinates and draw ( ) which draws them. Write a main ( ) program to exercise this class using graphic mode graphic functions.

---------- Cut here ----------

2. Write a C++ program to draw a straight line using simple DDA technique.

3. Draw a triangle ABC where A is (20,10), B is (30,50) and C is (40,10) Magnify the two dimensional object to the scaling factor of $Sx = 2, Sy = 2$.

---------- Cut here ----------

4. Draw a triangle ABC where A is (20,10), B is (30,50) and C is (40,10). Diminish the two dimensional object to the scaling factor of $Sx = 0.5, Sy = 0.5$.

---------- Cut here ----------

5. Draw a triangle ABC where A is (20,10), B is (30,50) and C is (40,10). Rotate the two dimensional object clockwise through the point origin at 30°.

---------- Cut here ----------

6. Draw a triangle ABC where A is (20,10), B is (30,50) and C is (40,10). Rotate the two dimensional object clockwise through an angle of 30° at the point A.

---------- Cut here ----------

7. Draw a triangle ABC where A is (20,10), B is (30,50) and C is (40,10). Translate the two dimensional object through the distance of $Tx = 10, Ty = 10$.

---------- Cut here ----------

8. Develop a program to do the following

   (a) Draw a triangle ABC, translate it to a distance and then rotate it through an angle clockwise.

   (b) With the triangle, reverse the operations as in case (a).

   (c) Check whether the resultant figures of case (a) and case (b) are one and the same or different.

9. Write a C++ program to manipulate the graphical object using SHADING technique.

---------- Cut here ----------

10. Implement the polygon clipping algorithm using C++ graphics mode graphics functions.

DE–8788 **26**

DISTANCE EDUCATION

P.G.D.C.A (S) EXAMINATION, MAY 2014.

Second Semester

UNIX AND SHELL PROGRAMMING – LAB IV

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

One question should be selected by the examiner and given to the candidates by LOT system.

Break-up of marks :

Record : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) A hospital maintains patient details. Write a shell program to list the following

   (i) Patients in a blood group wise.

   (ii) Patients in age between 20 to 30

   (iii) Exit.

   (b) Write a program for four-function calculator by using YACC command.

2. (a) Write a menu driven shell program for the following

   (i) List of files

   (ii) Processes of users

   (iii) Today's date

   (iv) Users of system

   (v) Quit to UNIX.

   (b) Write an execute a shell program to generate leave status report of an organization.

---------- Cut here ----------

3. (a) Write a shell program which accept the name of a file from the standard input and then perform the following tests on it

   (i) Enter the 5 names in a file

   (ii) Sort the names in existing file

   (iii) List unsorted and sorted file

   (iv) Quit.

   (b) Write an execute a shell program to print monthly pay slips for employees in an organisation. Assume your own data and generate accordingly.

---------- Cut here ----------

4. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it.

   (i) File existence

   (ii) File readable

   (iii) File writable

   (iv) Both readable and writable.

(b) Write a menu driven shell program to perform the following tasks

(i) Enter the sentence in a file.

(ii) Search a given whole word in an existing file.

(iii) Quit.

---------- Cut here ----------

5. (a) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located (using UNTIL statement)

(b) Write a menu driven shell program to copy, edit, rename, delete a file.